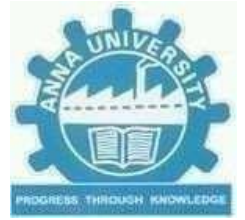




STUDENT DIGITAL PORTFOLIO & CREDENTIAL VERIFICATION SYSTEM USING BLOCKCHAIN



MINI PROJECT REPORT

Submitted by

ELAKKIYA M (6113231041030)

JANANI SRI M (6113231041050)

JAYADHARSHINI S (6113231041052)

In partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

Department of Computer Science and Engineering

MAHENDRA ENGINEERING COLLEGE

(Autonomous)

Mahendhirapuri, Mallasamudram, Namakkal Dt - 637 503

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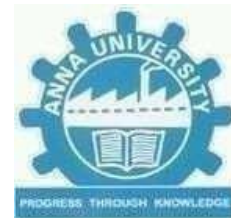
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MAY 2026



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BONAFIDE CERTIFICATE

Certified that this report titled **“STUDENT DIGITAL PORTFOLIO AND CREDENTIAL VERIFICATION USING BLOCKCHAIN WITH QR ACCESS”** is the Bonafide work of **ELAKKIYA M (Reg.no.6113231041030), JANANI SRI M (Reg.no.6113231041052), JAYADHARSHINI S (Reg.no.6113231041052)** who carried out the work under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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MAHENDRA ENGINEERING COLLEGE
(AUTONOMOUS)

INSTITUTION

VISION

- To be an internationally recognized institute for engineering education and research with ethical values.

MISSION

- To ensure the effective use of resources to mould the Students as professionals and entrepreneurs.
- To enhance the industry-institute interaction for Innovate technology practice.
- To encourage the faculty members and Students for advanced research.
- To inculcate ethical values among the faculty members and Students.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VISION

- To produce competent computer engineers proficient with state of the art technologies

MISSION

- To impart good quality technical education through an effective teaching-learning process.
- To enhance the students employability through mentoring and skill
- To encourage the faculty members and Students for advanced research.
- To inculcate ethical values among the faculty members and Students.

PROGRAM OUTCOMES(POs)

At the time of graduation, students from the Computer Science and Engineering program will possess: Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modelling to complex engineering problems with an understanding of the limitations. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for
i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES(PEOs)

The graduates of Computer Science and Engineering will be able to,

PEO1-Good communication, leadership and entrepreneurship skills

PEO2-Expertise on advanced computer technologies to become competitive

PEO3– The habit of learning and nurture there search attitude

PEO4– The ability to work in Steam with professional ethics

PROGRAM SPECIFIC OUTCOMES(PSOs)

Engineering Graduates will be able to,

PSO1 - Ability to comprehend the underlying principles and systematic methods for the development, operation, and maintenance of software, using professional engineering practices.

PSO2 - Ability to develop socially acceptable technical solutions to real world problems with various strategies for sustainable development.

PSO3 - Ability to apply the skills in the areas related to Algorithms, Networking, Web Designing, Artificial Intelligence, Internet of Things and Data Analytics of various complexities towards successful employment

PO ATTAINMENT

KEY WORDS	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
BLOCKCHAIN		✓	✓	✓	✓			✓		✓	✓		✓	✓
SMART CONTRACT	✓	✓	✓	✓		✓		✓	✓		✓	✓	✓	✓
IPFS STORAGE	✓	✓		✓	✓		✓	✓	✓	✓	✓	✓	✓	✓
QR VERIFY	✓		✓			✓	✓		✓	✓	✓	✓	✓	✓
SHA 256	✓		✓			✓	✓		✓	✓	✓	✓		✓

ABSTRACT

Blockchain technology is transforming digital ecosystems by enabling secure, tamper-proof, and trustless verification of academic credentials in real time. The proposed Student Digital Portfolio and Credential Verification System leverages decentralized technologies to overcome the limitations of traditional verification methods, which are often slow, centralized, and vulnerable to forgery. The system integrates Polygon blockchain, InterPlanetary File System (IPFS), SHA-256 hashing, and QR code-based verification to ensure authenticity, integrity, and transparency of academic records. Certificates are securely stored in IPFS, while their cryptographic hashes are permanently recorded on the blockchain, providing immutable proof of ownership and preventing unauthorized modifications. A multi-signature approval mechanism involving authorized faculty members further strengthens credential validation before blockchain registration. Developed using technologies such as Hardhat, MongoDB Atlas, Vercel, and Railway, and deployed on the Polygon Mumbai testnet, the system offers scalable, low-cost, and reliable performance for modern educational and recruitment environments. By combining decentralized storage, blockchain security, and real-time verification, the proposed framework establishes a transparent, efficient, and highly secure solution for digital academic credential management and verification.

(Keywords: Blockchain, Credential Verification, IPFS, Smart Contracts, QR Code Authentication, Digital Portfolio, Polygon Network, Decentralized Storage).

சுருக்கம்

Blockchain தொழில்நுட்பம், கல்வி சான்றிதழ்களின் பாதுகாப்பான, மாற்றமற்ற மற்றும் நம்பகமான நிகழ்நேர சரிபார்ப்பை வழங்குவதன் மூலம் டிஜிட்டல் சூழல்களை மாற்றி அமைத்து வருகிறது. முன்மொழியப்பட்ட *Student Digital Portfolio and Credential Verification System* என்பது மையமற்ற (decentralized) தொழில்நுட்பங்களை பயன்படுத்தி, தற்போதைய சான்றிதழ் சரிபார்ப்பு முறைகளில் உள்ள மந்தநிலை, மையப்படுத்தப்பட்ட கட்டுப்பாடு மற்றும் போலி ஆவண அபாயங்களை தீர்க்கும் வகையில் உருவாக்கப்பட்டுள்ளது. இந்த அமைப்பில் Polygon blockchain, InterPlanetary File System (IPFS), SHA-256 hashing மற்றும் QR code அடிப்படையிலான சரிபார்ப்பு ஆகியவை ஒருங்கிணைக்கப்பட்டு, கல்வி பதிவுகளின் நம்பகத்தன்மை, முழுமைத்தன்மை மற்றும் வெளிப்படைத்தன்மை உறுதி செய்யப்படுகின்றன. சான்றிதழ்கள் IPFS-ல் பாதுகாப்பாக சேமிக்கப்படுகின்றன; அவற்றின் குறியாக்க ஹாஷ் மதிப்புகள் blockchain-ல் நிரந்தரமாக பதிவு செய்யப்படுவதால், தரவு மாற்றம் மற்றும் அங்கீகரிக்கப்படாத திருத்தங்கள் தடுக்கப்படுகின்றன. மேலும், அங்கீகரிக்கப்பட்ட பேராசிரியர்கள் பங்கேற்கும் multi-signature approval முறையின் மூலம் சான்றிதழ்களின் உண்மைத்தன்மை blockchain பதிவிற்கு முன் உறுதிப்படுத்தப்படுகிறது. Hardhat, MongoDB Atlas, Vercel மற்றும் Railway போன்ற நவீன தொழில்நுட்பங்களை பயன்படுத்தி உருவாக்கப்பட்ட இந்த அமைப்பு Polygon Mumbai testnet-ல் செயல்படுத்தப்பட்டு, அளவீட்டுத் திறன், குறைந்த செலவு மற்றும் நம்பகமான செயல்திறனை வழங்குகிறது. மையமற்ற சேமிப்பு, blockchain பாதுகாப்பு மற்றும் நிகழ்நேர சரிபார்ப்பை ஒருங்கிணைப்பதன் மூலம், இந்த அமைப்பு கல்வி சான்றிதழ் மேலாண்மை மற்றும் சரிபார்ப்பிற்கான பாதுகாப்பான, திறமையான மற்றும் வெளிப்படையான தீர்வாக திகழ்கிறது.

(முக்கிய சொற்கள்: Blockchain, சான்றிதழ் சரிபார்ப்பு, IPFS, Smart Contracts, QR குறியீட்டு அங்கீகாரம், டிஜிட்டல் போர்ட்: போலியோ, Polygon Network, மையமற்ற சேமிப்பு).

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LIST OF ABBREVIATIONS

API	Application Programming Interface
CSS	Cascading Style Sheet
DB	Dashbase
HTTP	Hypertext Transfer Protocol
IPFS	Inter Planetary File System
PDF	Portable Document File
QR	Quick Response
SHA	Secure HAsH Algorithm
SQL	Structured Query Language
UI	User Interface

CHAPTER- 1

INTRODUCTION

Blockchain technology is revolutionizing digital ecosystems by enabling secure, tamper-proof, and trustless verification of academic credentials in real time. As education and professional systems increasingly transition toward digital platforms, the generation, storage, and sharing of academic certificates have become fully digitized processes. These credentials serve as critical proof of knowledge, skills, and qualifications in recruitment, higher education, and professional validation. However, the growing reliance on digital credentials has also exposed significant challenges related to authenticity, security, and trust, particularly in environments where verification processes are still dependent on manual or centralized systems.

Traditional credential verification methods involve multiple administrative steps, including institutional validation, email confirmations, and database cross-checking. These processes are not only time-consuming but also inefficient, often resulting in delays ranging from several days to weeks. Moreover, centralized systems introduce limitations such as lack of interoperability, restricted accessibility, and vulnerability to data breaches or manipulation. In global scenarios, where credentials must be verified across multiple institutions and regions, the absence of a standardized and trustworthy verification mechanism further complicates the process. These limitations create bottlenecks in recruitment and academic evaluation, affecting both organizations and individuals.

To address these challenges, the project **“Student Digital Portfolio & Credential Verification System using blockchain”** introduces a decentralized framework that leverages blockchain technology for secure and transparent credential management. Instead of relying on a central authority, the system utilizes the **Polygon blockchain (Mumbai testnet)** to record cryptographic representations of certificates. Each credential is processed using SHA-256 hashing, and the generated hash is stored on the blockchain, ensuring immutability and resistance to tampering. Once recorded, the data cannot be altered, providing a reliable and trustless verification mechanism that eliminates the need for intermediaries.

While blockchain ensures data integrity and immutability, storing large certificate files directly on-chain is not efficient. To overcome this limitation, the system integrates the **Inter Planetary File System (IPFS)**, a decentralized storage protocol that distributes files across a peer-to-peer network. Each certificate is stored securely in IPFS and is associated with a unique content identifier, which is linked to its corresponding blockchain record. This combination of blockchain and IPFS ensures both security and scalability, allowing efficient storage and retrieval of credential data without compromising performance.

The proposed system consists of several key modules that work together to ensure secure, decentralized, and efficient credential management. Each module is responsible for a specific function within the system, enabling a structured and scalable workflow from certificate submission to verification.

Core Modules:

1. **Certificate Upload Module** – Enables students to upload academic credentials through a user-friendly interface.
2. **Hash Generation Module** – Generates a unique SHA-256 hash for each certificate to ensure data integrity.
3. **Blockchain Storage Module** – Records the hash value on the Polygon blockchain using smart contracts, ensuring immutability.
4. **IPFS Storage Module** – Stores the actual certificate file in a decentralized manner using IPFS.
5. **Multi-Signature Validation Module** – Ensures approval from multiple faculty members before final validation.
6. **QR Code Generation & Verification Module** – Generates QR codes for each credential and enables instant verification.

From a technological perspective, this project represents the integration of blockchain, decentralized storage, and real-time verification mechanisms, forming a robust framework for modern credential management. By shifting from traditional centralized models to a decentralized trust-based architecture, the system enhances security, reduces operational delays, and improves overall efficiency. This transition plays a crucial role in addressing the evolving demands of digital education and global employment systems.

In conclusion, the proposed system provides a secure, scalable, and efficient solution for credential verification by combining blockchain immutability, IPFS-based storage, and QR-enabled instant access. It eliminates the limitations of traditional verification methods, reduces the risk of fraud, and establishes a reliable framework for trustless credential validation. Ultimately, the system contributes to building a future where academic credentials are not only digital but also universally verifiable, transparent, and secure.

1.1 OBJECTIVES

With the rapid digitalization of education and recruitment systems, academic credentials are increasingly generated, shared, and verified in electronic formats. However, existing verification mechanisms rely heavily on manual processes and centralized databases, leading to delays, lack of transparency, and vulnerability to certificate forgery. These limitations create significant challenges for institutions and employers, especially in global scenarios where trust and instant validation are critical. As a result, there is a growing need for a **secure, decentralized, and efficient system** that ensures authenticity and enables real-time verification of academic credentials.

The present study, titled “**Student Digital Portfolio & Credential Verification System Using Blockchain**”, aims to design and develop a framework that leverages **blockchain technology, IPFS, and QR code-based verification** to address these challenges. The system focuses on creating a tamper-proof and trustless environment where credentials can be securely stored, managed, and verified without relying on centralized authorities. By integrating decentralized storage and cryptographic validation, the proposed system enhances both security and efficiency in credential management.

The main goal of the present study is to **eliminate delays, prevent credential forgery, and establish a transparent verification framework** through the use of decentralized technologies. The project emphasizes real-time accessibility, data integrity, and minimal manual intervention, ensuring that credential verification becomes faster, more reliable, and universally accessible across institutions and organizations.

1.1 To develop a blockchain-based credential verification system

The primary objective is to design a system that records cryptographic hashes of academic certificates on the blockchain. By utilizing secure hashing algorithms such as SHA-256 and storing the results on the Polygon blockchain, the system ensures immutability and prevents unauthorized modifications. This enables a trustless verification mechanism where credentials can be validated without relying on intermediaries.

This objective focuses on enhancing security and transparency while eliminating the risks associated with centralized data storage.

1.2 To implement decentralized storage using IPFS

Another key objective is to integrate the InterPlanetary File System (IPFS) for storing certificate data in a distributed manner. Instead of relying on centralized servers, the system stores files across a peer-to-peer network and assigns unique content identifiers for retrieval.

Additionally, integrating IPFS with blockchain ensures that while the actual files are stored off-chain for efficiency, their integrity is verified through on-chain hash references. This ensures high availability, scalability, and protection against data loss or tampering, while improving overall system efficiency.

1.3 To enable real-time verification using QR code integration

The project also aims to include on-device continual learning, where deployed models can adapt and improve over time using real-world data. This objective ensures that the system remains intelligent and self-evolving without requiring frequent reprogramming.

The system ensures that the retrieved data matches the original blockchain-stored hash, thereby confirming its integrity and preventing the acceptance of forged documents. By reducing verification time from days to seconds, this objective significantly improves efficiency, enhances user convenience, and enables seamless credential validation.

1. To incorporate a multi-signature validation mechanism

A crucial objective of the system is to implement a **multi-signature approval mechanism** to ensure that only authenticated and verified credentials are recorded on the blockchain. In this approach, a certificate must be approved by multiple authorized faculty members or institutional authorities before it is considered valid and permanently stored.

This enhances trust and ensures that only authenticated and approved credentials are stored, preventing unauthorized issuance and strengthening the overall validation process.

2.To design a user-friendly digital portfolio system

The final objective is to design and develop a **comprehensive digital portfolio platform** that allows students to securely upload, manage, and share their academic credentials. The system provides an intuitive interface where users can organize their certificates, track their verification status, and generate QR codes for easy sharing with employers or institutions.

The platform is designed with a focus on usability and accessibility, ensuring that even non-technical users can interact with the system without difficulty. At the same time, it incorporates strong backend security measures, including authentication mechanisms and encrypted data handling, to protect sensitive information.

The platform is designed with a focus on usability and accessibility, ensuring that even non-technical users can interact with the system without difficulty. At the same time, it incorporates strong backend security measures, including authentication mechanisms and encrypted data handling, to protect sensitive information.

CHAPTER-2

LITERATURE SURVEY

2.1 BLOCKCHAIN-BASED ACADEMIC CREDENTIALS VERIFICATION SYSTEMS

Kumar, A., & Singh, R. (2024). *Blockchain-Based Academic Credential Verification System*. IEEE Access.

This study presents a blockchain-based framework for issuing and verifying academic credentials using decentralized ledgers. The system ensures immutability and transparency by storing cryptographic hashes of certificates on the blockchain. It eliminates the need for third-party verification and reduces the risk of certificate forgery. The authors highlight the advantages of decentralization in improving trust and reducing verification time.

However, the system primarily focuses on blockchain storage and does not address efficient handling of large certificate files or decentralized storage mechanisms. The proposed system improves this by integrating IPFS for scalable storage and combining it with real-time QR-based verification.

2.2 BLOCKCHAIN AND QR CODE-BASED VERIFICATION SYSTEMS

Zhang, Y., et al. (2025). *Secure Document Verification using Blockchain and QR Codes*. Springer Journal of Information Security.

This paper explores the integration of blockchain with QR code technology for secure document verification. The system links QR codes to blockchain records, enabling instant access to credential data and verification status. The approach significantly reduces verification delays and enhances user convenience.

While the system provides efficient access and verification, it lacks a decentralized storage mechanism for handling large files and does not incorporate multi-level validation processes. The proposed system extends this by integrating IPFS storage and multi-signature validation for improved security and scalability.

2.3 DECENTRALIZED STORAGE USING IPFS

Xubin Wang, & Weijia Jia (2025). “Optimizing Edge AI: A Comprehensive Survey on Data, Model, and System Strategies,” arXiv preprint arXiv:2501.03265.

This paper delivers a comprehensive evaluation of edge computing platforms for AI workloads, focusing on computational efficiency, power consumption, and hardware compatibility. It introduces a tri-level optimization framework that considers data handling, model compression, and system-level design as interconnected factors influencing AI performance at the edge. Through detailed analysis, the authors assess multiple platforms, highlighting the trade-offs between energy efficiency, inference speed, and thermal constraints. The study emphasizes techniques such as pruning, quantization, and hardware acceleration to optimize deep learning models for real-time inference on resource-limited devices.

2.4 EXISTING DIGITAL CREDENTIAL SYSTEMS

Patel, S., & Mehta, R. (2025). *Digital Credential Systems: A Comparative Study of Centralized Platforms*. ACM Digital Library.

This section examines existing digital credential systems and their approaches to storing, managing, and verifying academic certificates through centralized platforms. The referenced study presents a comparative analysis of widely used systems such as DigiLocker, Certopus, and Truscholar, focusing on their architectural design, usability, and security mechanisms. These platforms enable digital storage and sharing of credentials, improving accessibility and reducing dependence on physical documents. The authors also highlight features such as institutional validation workflows and controlled access mechanisms that support credential management. The proposed system addresses these limitations by introducing a decentralized framework that leverages blockchain technology for immutable credential verification. By integrating IPFS for distributed storage and QR code-based real-time validation, the system establishes a seamless end-to-end verification process—from certificate upload and multi-signature approval to instant verification—thereby enhancing security, transparency, and operational efficiency across digital credential ecosystems.

2.5 SECURITY AND PRIVACY IN DIGITAL CREDENTIAL VERIFICATION SYSTEMS

Xubin Wang et al. (2025). “Empowering Edge Intelligence: A Comprehensive Survey on On-Device AI Models,” arXiv preprint arXiv:2501.04112.

This study investigates the implementation of AI models directly on resource-constrained devices, emphasizing lightweight architectures, adaptive learning, and low-latency inference. The authors provide an in-depth overview of model compression techniques and adaptive AI frameworks designed to operate efficiently within limited computational budgets. They analyze real-time decision-making capabilities and privacy-preserving mechanisms that minimize data transmission to the cloud. Evaluation metrics include inference accuracy, latency, and resource utilization. Despite significant progress, the need for flexible and adaptive systems capable of dynamically adjusting to new sensors and changing workloads remains unmet. Prompt2Edge contributes to this evolution by introducing self-configuring modules capable of detecting new peripherals and adapting AI models accordingly, ensuring continuous performance improvement without manual intervention.

CHAPTER-3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

In the current digital ecosystem, academic credential management and verification are primarily handled through centralized systems such as institutional databases, digital document repositories, and third-party verification platforms. Students typically receive certificates in digital or physical formats, which are then stored in centralized servers or personal storage systems. Verification of these credentials involves multiple administrative steps, including manual validation by institutions, email confirmations, or database cross-checking.

This process is often time-consuming and inefficient, with verification delays ranging from several days to weeks. In many cases, organizations must directly contact issuing institutions to confirm authenticity, leading to communication overhead and operational bottlenecks. Furthermore, these systems lack a standardized global framework, making cross-institutional and international verification even more complex.

Existing platforms such as DigiLocker, Certopus, and Truscholar provide digital storage and limited verification capabilities. While these systems improve accessibility and reduce reliance on physical documents, they are still dependent on centralized infrastructure. This introduces risks such as data breaches, unauthorized access, and single points of failure. Additionally, the absence of strong cryptographic validation mechanisms increases the possibility of certificate forgery and manipulation.

Another major limitation is the lack of real-time verification. Most systems do not support instant validation, requiring manual intervention or platform-specific access. Moreover, users have limited control over their credentials, as data ownership remains with institutions or service providers rather than the individuals themselves. Hence, the current credential verification ecosystem lacks **transparency, security, scalability, and trust**, highlighting the need for a more robust and decentralized solution.

3.1.1 DISADVANTAGES

The limitations of the existing systems can be summarized as follows:

1. **Centralized Architecture:** Most systems rely on a single authority, making them vulnerable to data breaches and system failures.
2. **Time-Consuming Verification:** Manual processes and institutional dependencies lead to delays ranging from days to weeks.
3. **Risk of Forgery:** Lack of strong cryptographic validation allows manipulation or duplication of certificates.
4. **Limited Interoperability:** Systems are not universally compatible across institutions and regions.
5. **Lack of Real-Time Access:** Verification is not instantaneous and often requires manual approval.
6. **Data Ownership Issues:** Users do not have full control over their credentials, leading to privacy concerns.
7. **Scalability Challenges:** Handling large volumes of credential data efficiently remains difficult.

3.2 PROPOSED SYSTEM

The proposed system, “**Student Digital Portfolio & Credential Verification System using Blockchain, IPFS, and QR Code,**” introduces a decentralized and secure framework for managing and verifying academic credentials.

Unlike traditional systems that depend on centralized authorities, the proposed solution leverages **blockchain technology** to store cryptographic hashes of certificates, ensuring immutability and tamper-proof verification. Each credential is processed using **SHA-256 hashing**, and the generated hash is recorded on the Polygon blockchain (Mumbai testnet), providing a transparent and trustless validation mechanism.

To address storage limitations, the system integrates the **InterPlanetary File System (IPFS)** for decentralized file storage. Certificates are stored off-chain in IPFS and linked to blockchain records through unique content identifiers, ensuring efficient storage and secure retrieval without compromising performance.

The system also incorporates a **multi-signature validation mechanism**, where credentials must be approved by multiple authorized faculty members before being recorded.

For real-time verification, the system generates **QR codes** linked to credential data. Employers and institutions can instantly verify certificates by scanning the QR code, enabling fast and seamless validation without intermediary involvement.

Additionally, the platform provides a **digital portfolio interface** that allows students to upload, manage, and share their verified credentials securely. The system is designed to be scalable, user-friendly, and globally accessible, ensuring efficient interaction between students, institutions, and employers.

The proposed system integrates multiple advanced technologies to ensure secure, efficient, and real-time credential management and verification:

1. **Blockchain Technology (Polygon Network)** for storing cryptographic hashes of certificates, ensuring immutability, transparency, and tamper-proof validation.
2. **SHA-256 Hashing Mechanism** for generating unique digital fingerprints of certificates, enabling secure identity representation without exposing actual data.
3. **Inter Planetary File System (IPFS)** for decentralized storage of certificate files, ensuring scalability, high availability, and resistance to data loss or manipulation.
4. **Multi-Signature Validation System** for verifying credentials through approval from multiple authorized faculty members before recording them on the blockchain.
5. **QR Code-Based Verification** for enabling instant and real-time access to credential authenticity through simple scanning.
6. **Digital Portfolio Interface** for allowing students to upload, manage, and securely share their verified credentials with institutions and employers.

This system eliminates the dependency on centralized authorities and manual verification processes, enabling a **trustless and automated credential validation mechanism**. Its modular and decentralized architecture ensures scalability, security, and global accessibility, making it suitable for diverse academic and professional ecosystems.

3.2.1 ADVANTAGES

The proposed system offers several significant advantages over existing methods:

1. **Decentralized Verification:** Eliminates dependency on central authorities, ensuring transparency and trust.
2. **Tamper-Proof Security:** Blockchain ensures that once recorded, credentials cannot be altered or forged.
3. **Real-Time Verification:** QR-based access enables instant validation within seconds.
4. **Enhanced Data Integrity:** SHA-256 hashing guarantees authenticity of stored credentials.
5. **Secure Storage:** IPFS provides decentralized and scalable storage for certificate data.
6. **Multi-Signature Approval:** Ensures that only verified and authorized credentials are recorded.
7. **User Ownership:** Students have full control over their digital credentials and sharing permissions.
8. **Scalability and Efficiency:** Supports large-scale deployment without performance degradation.
9. **Reduced Verification Time:** Eliminates delays associated with manual processes.

CHAPTER- 4

SYSTEM REQUIREMENTS

4.1 HARDWARE REQUIRED

The Student Digital Portfolio & Credential Verification System utilizes standard computing and mobile hardware components to support secure credential storage, blockchain interaction, QR code generation, and verification processes. The hardware setup is designed to be cost-effective, scalable, and suitable for web-based as well as mobile-based verification systems. The system mainly depends on cloud technologies such as Polygon Blockchain, IPFS, and MongoDB Atlas, reducing the need for expensive dedicated hardware.

1. Edge Compute Unit (Microcontroller / Single Board Computer)

- **Component:** Android Smartphone / iPhone.
- **Purpose:** Acts as the primary development and testing platform for frontend, backend, blockchain integration, and database management.
- **Features:** Minimum 8 GB RAM for smooth multitasking, 256 GB or higher SSD storage, Supports Node.js, React.js, MongoDB, and blockchain tools, Internet connectivity for cloud and blockchain access.

2. Sensor and Peripheral Modules

- **Camera Module (Mobile Camera / Webcam):** Used for scanning QR codes and verifying digital portfolios instantly through mobile or web applications.
- **QR Code Scanner:** Enables quick retrieval of blockchain transaction details linked with student credentials.
- **Display Device (Monitor / Laptop Screen):** Used for displaying dashboards, portfolio details, approval status, and verification results.
- **Networking Module (Wi-Fi / Ethernet):** Provides internet connectivity for accessing Polygon Blockchain, IPFS storage, and cloud databases.
- **Storage Devices (SSD / HDD):** Used for storing project files, backend services, databases, and generated portfolio documents.

3. Power Supply

- **Laptop / PC Power Adapter:** Supplies power to the development system used for frontend, backend, blockchain, and database operations.
- **Mobile Device Battery / Charger:** Provides power for smartphones used in QR code scanning and mobile verification testing.
- **Purpose:** Supports mobile devices used for QR verification
- Maintains stable internet and blockchain connectivity
- **Features:** Stable and uninterrupted power delivery, Supports long-duration system operation.

4. Storage Devices

- **Component:** SSD (Solid State Drive) / HDD (Hard Disk Drive)
- **Purpose:** Used for storing project files, databases, certificates, portfolio documents, and software tools required for development and verification.
- **Features:** High-speed data storage and retrieval, Reliable storage for large datasets, Supports backup and recovery.

5. Communication Modules

- **Component:** Wi-Fi Module / Ethernet Network Adapter.
- **Purpose:** Used to establish internet connectivity for accessing Polygon Blockchain, IPFS storage, cloud databases, and online verification services.
- **Features:** Supports high-speed data communication, Enables real-time blockchain interaction Provides stable internet connectivity.

6. Networking and Cloud Connectivity

- **Component:** Wi-Fi Router / Ethernet Network Adapter
- **Purpose:** Used to provide internet and network connectivity for accessing blockchain services, cloud databases, IPFS storage, and online credential verification.
- **Features:** High-speed network communication, Stable internet connectivity, Supports real-time data transfer.

4.2 SOFTWARE REQUIRED

The “Student Digital Portfolio & Credential Verification System” uses blockchain, IPFS, QR codes, and web technologies for secure credential storage, verification, and system deployment.

1. Operating System

- **Component:** Windows / Linux (Ubuntu)
- **Purpose:** Provides the development and execution environment for frontend backend integration and blockchain deployment
- **Features:** Multi-user support, command-line integration, stable performance, and compatibility with Web3 development environments.

2. Programming Languages

- **Component:** JavaScript, Solidity, HTML, CSS
- **Purpose:**
 - a. **JavaScript** is used for frontend interaction, backend services, and blockchain communication.
 - b. **Solidity** is used for developing and deploying smart contracts on the Polygon blockchain.
 - c. **HTML and CSS** are used for designing responsive user interfaces and digital portfolio pages.
- **Features:** Cross-platform compatibility, efficient blockchain integration, responsive UI development, and extensive Web3 ecosystem support.

3. Frameworks and Libraries

- **Hardhat:** Used for smart contract development, testing, debugging, and deployment on the Polygon Mumbai testnet.
- **Ethers.js / Web3.js:** Enables communication between the web application and blockchain smart contracts.
- **SHA-256 Cryptographic Hashing:** Generates secure digital fingerprints of certificates before blockchain storage.
- **MetaMask Wallet:** Facilitates blockchain authentication and wallet connectivity

4. Decentralized Storage Integration

- **Platform:** InterPlanetary File System (IPFS)
- **Purpose:** Stores certificate files in a decentralized peer-to-peer storage network instead of centralized servers.
- **Features:** Distributed storage, high availability, content-based addressing, scalability, and protection against data tampering or data loss.

5. Development Environment

- **Component:** Visual Studio Code (VS Code)
- **Purpose:** Provides an integrated environment for developing frontend interfaces, backend APIs, blockchain smart contracts, and deployment scripts..
- **Features:** Lightweight architecture, debugging tools, extension support, Git integration, and efficient project management capabilities.

6. Database

- **Component:** MongoDB Atlas
- **Purpose:** Stores user information, portfolio metadata, verification history, and system-related records required for application functionality.
- **Features:** Cloud-based storage, scalability, real-time access, flexible document structure, and secure database management.

7. Cloud Hosting and Deployment Platforms

- **Component:** Vercel / Railway
- **Purpose:**
 - a. **Vercel** is used for hosting the frontend React application.
 - b. **Railway** is used for backend deployment and API hosting.
- **Features:** Lightweight, fast, and suitable for constrained IoT networks.

8. Frontend and Backend Technologies

- **Component:** React.js / Node.js / Express.js
- **Purpose:**
 - a. **React.js** provides an interactive and responsive digital portfolio interface for students and verifiers.
 - b. **Node.js and Express.js** handle backend logic, API management, QR generation, and blockchain communication.
- **Features:** Fast rendering, scalable architecture, responsive design, REST API integration, and real-time interaction.

QR Code Generation and Verification Tools

- **Component:** QR Code Generator Libraries
- **Purpose:** Generates unique QR codes linked to blockchain-verified credentials for instant authentication and validation.
- **Features:** Real-time verification, mobile accessibility, quick response generation, and simplified credential sharing.

Version Control and Collaboration Tools

- **Component:** Git and GitHub
- **Purpose:** Used for source code management, version tracking, and collaborative development of the project.
- **Features:** Distributed version control, secure repository management, rollback support, and team collaboration.

CHAPTER- 5

SYSTEM DESIGN

5.1 ARCHITECTURE DIAGRAM

The **Student Digital Portfolio & Credential Verification System** uses a five-layer distributed architecture to provide secure, tamper-proof, and decentralized academic management.

1. USER INTERACTION LAYER

- **INTERFACES:** A React-based Progressive Web App (PWA) for Students (portfolio management), Faculty (approval dashboards), and Verifiers (employers).
- **FUNCTION:** Allows credential uploads, role-based reviews, and QR-enabled mobile verification via secure REST APIs.

2. APPLICATION & SERVICE LAYER

- **ORCHESTRATION:** Manages authentication, QR code generation, and communicates with the blockchain via Web3.js.

3. BLOCKCHAIN PROCESSING LAYER

- **IMMUTABILITY:** Stores cryptographic hashes and metadata (not the full file) to ensure records are permanent, auditable, and forgery-proof.
- **AUTOMATION:** Handles issuance, multi-signature validation, and revocation.

4. DECENTRALIZED STORAGE LAYER

- **IPFS:** Stores actual certificate files using Content Identifiers (CIDs). This ensures decentralized, fault-tolerant access without a central server.
- **MONGODB ATLAS:** A hybrid cloud database used to store application metadata, user profiles, and verification logs.

5. VERIFICATION & ANALYTICS LAYER

- **PROCESS:** Verifiers scan a QR Code to trigger a real-time integrity check. The system re-hashes the IPFS file and matches it against the Blockchain record.
- **AUDIT:** Tracks verification history and provides analytics on credential usage and institutional activity.

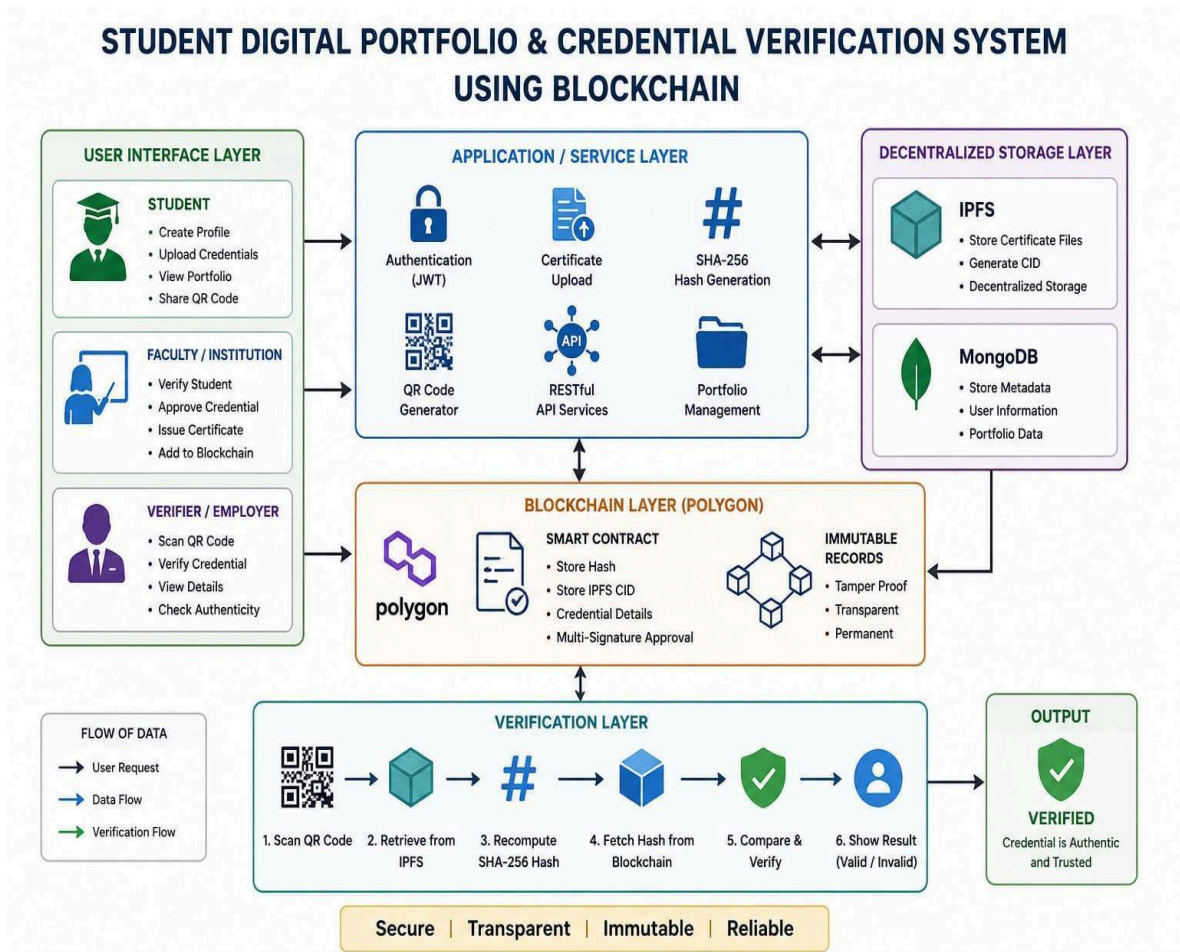


Figure 5.1: Platform Architecture

5.2 DATA FLOW DIAGRAM (DFD)

The Data Flow Diagram illustrates how academic records move from the student to the blockchain and eventually to the verifier.

LEVEL 0: CONTEXT DIAGRAM

- **STUDENT:** Sends credential data and profile info to the system; receives verification status and digital portfolio access.
- **FACULTY:** Receives pending credentials for review; sends approval/rejection signals to the system.
- **VERIFIER:** Sends QR scan/verification requests; receives validated credential data and authenticity reports.

LEVEL 1: PROCESS FLOW

- **REGISTRATION & AUTHENTICATION:** Users (Students/Faculty) register and connect their Blockchain Wallets.
- **CREDENTIAL UPLOAD & HASHING:** Student uploads a document. The system generates a SHA-256 Hash.
- **VALIDATION WORKFLOW:** Faculty reviews the document. Upon approval, the system initiates a blockchain transaction.
- **DECENTRALIZED STORAGE:** The file is pushed to IPFS (receives a CID), and the CID + Hash are recorded on the Polygon Blockchain.
- **QR GENERATION:** A unique QR code is generated linking the IPFS file and the Blockchain transaction ID.
- **VERIFICATION:** Verifier scans the QR. The system fetches the file from IPFS, re-hashes it, and compares it with the Blockchain record.

5.3 DATABASE DESIGN (SCHEMA)

The system uses a hybrid storage approach. Sensitive documents are stored on IPFS, while application metadata is managed in MongoDB Atlas.

1. User Collection (Metadata)

FIELD NAME	DATA TYPE	DESCRIPTION
user_id	ObjectID	Unique identifier for the user
full_name	String	Legal name of the student/faculty
email	String	Registered email address
role	String	Student, Faculty, or Admin
wallet_addr	String	Polygon blockchain wallet address

2. Credentials Collection (IPFS & Blockchain Ref)

FIELD NAME	DATA TYPE	DESCRIPTION
cred_id	ObjectID	Unique identifier for the credential
student_id	ObjectID	Reference to the student
file_name	String	Name of the uploaded document
ipfs_cid	String	Content Identifier from IPFS storage
file_hash	String	SHA-256 hash of the original file
tx_hash	String	Blockchain transaction hash (Polygon)
status	String	Pending, Approved, or Rejected

3. Verification Logs Collection

FIELD NAME	DATA TYPE	DESCRIPTION
log_id	ObjectID	Unique ID for the verification event
cred_id	ObjectID	Reference to the credential verified
verifier_id	String	Identifier of the external organization
timestamp	DateTime	Date and time of the QR scan
result	Boolean	Authenticity status (Valid/Invalid)

4. Portfolio Collection

FIELD NAME	DATA TYPE	DESCRIPTION
portfolio_id	ObjectID	Unique ID for the student portfolio
student_id	ObjectID	Reference to the student
skills	Array	List of verified skills
project_links	Array	External links to student projects

CHAPTER- 6

SYSTEM IMPLEMENTATION

6.1 CERTIFICATE UPLOAD MODULE

The Certificate Upload Module acts as the primary entry point of the system, enabling students to upload academic certificates, project documents, and portfolio files through a responsive web-based interface. The module supports secure drag-and-drop file upload functionality and validates document formats before processing. It establishes seamless interaction between the frontend interface and backend services, ensuring efficient credential submission and portfolio management.

6.2 HASH GENERATION MODULE

The Hash Generation Module is responsible for generating a unique SHA-256 cryptographic hash for every uploaded certificate. The generated hash acts as a digital fingerprint of the document, ensuring data integrity and tamper detection. Even a minor modification in the uploaded file produces a completely different hash value, enabling reliable verification of document authenticity before blockchain registration.

6.3 BLOCKCHAIN STORAGE MODULE

The Blockchain Storage Module records credential hash values and verification metadata on the Polygon blockchain using Solidity smart contracts. Instead of storing complete certificate files directly on-chain, the system stores immutable cryptographic references and transaction details to improve scalability and reduce blockchain storage overhead. This module ensures transparency, immutability, and tamper-proof credential validation within the verification ecosystem.

6.4 IPFS STORAGE MODULE

The IPFS Storage Module securely stores uploaded certificates and portfolio documents in a decentralized manner using IPFS. Each uploaded file receives a unique Content Identifier (CID), ensuring secure access, data integrity, and protection from unauthorized modification or server failure.

6.5 MULTI-SIGNATURE VALIDATION MODULE

The Multi-Signature Validation Module introduces institutional approval mechanisms into the credential issuance workflow. Before a credential is finalized on the blockchain, approval from multiple authorized faculty members is required through smart contract–based validation. This multi-signature architecture enhances trust, prevents unauthorized credential issuance, and strengthens institutional accountability within the verification process.

6.6 QR CODE GENERATION AND VERIFICATION MODULE

The QR Code Generation and Verification Module generates unique QR codes linked to blockchain transaction records and IPFS document references. The generated QR code is embedded into the digital credential and enables instant verification through mobile devices or web interfaces. During verification, the system retrieves blockchain data and recomputes the document hash to validate authenticity, ensuring rapid and secure credential verification without requiring manual institutional confirmation.

6.7 USER AUTHENTICATION AND ACCESS CONTROL MODULE

The User Authentication and Access Control Module manages secure login, identity verification, and role-based system authorization for students, faculty members, and external verifiers. The module integrates MetaMask wallet authentication and JWT-based session management to ensure secure access to system resources and blockchain operations.

Role-based access policies restrict unauthorized actions by assigning different privileges to users such as certificate upload, approval, verification, and audit monitoring. This module enhances system security, protects sensitive academic data, and ensures controlled interaction within the decentralized credential verification ecosystem.

CHAPTER-7

SYSTEM TESTING

7.1 TYPES OF TESTING PERFORMED

System testing for the proposed credential verification system was conducted through multiple testing phases to validate the correctness, integration, security, and usability of all system components .

7.1.1 FUNCTIONAL TESTING

Functional testing was performed to verify that all core modules of the system operated according to the specified requirements. The testing process validated functionalities such as certificate upload, SHA-256 hash generation, IPFS storage, blockchain transaction execution, faculty approval workflows, QR code generation, and credential verification. Both valid and invalid input scenarios were tested to ensure proper error handling and reliable system behavior during credential processing and verification operations.

7.1.2 INTEGRATION TESTING

Integration testing was conducted to ensure seamless communication and data flow between the frontend interface, backend services, blockchain network, IPFS storage, and verification modules. The testing validated successful interaction between React-based user interfaces, Node.js APIs, Solidity smart contracts, and decentralized storage services. It also confirmed that blockchain records, IPFS content identifiers, and QR verification data remained properly synchronized throughout the credential lifecycle.

7.1.3 PERFORMANCE TESTING

Performance testing evaluated the responsiveness, scalability, and operational efficiency of the system under different workloads. Parameters such as certificate upload time, blockchain transaction latency, IPFS storage speed, and QR verification response time were measured and analyzed. Stress testing was additionally performed to validate system stability during multiple concurrent uploads and verification requests, ensuring consistent performance in real-time academic environments.

7.1.4 SECURITY TESTING

Security testing was crucial to ensure that all processes—from certificate upload to blockchain verification—were protected against unauthorized access and data breaches. Tests were conducted to verify secure communication between the application, blockchain network, IPFS storage, and database services. Vulnerability scans identified risks such as unsecured APIs, unauthorized access, and credential manipulation, which were then mitigated. This ensured the confidentiality, integrity, and authenticity of all credentials managed by the system.

7.1.5 USER ACCEPTANCE TESTING (UAT)

User Acceptance Testing was conducted to evaluate the usability, reliability, and practical functionality of the system from the perspective of students, faculty members, and employers. Users tested certificate upload, approval workflows, QR-based verification, and portfolio management features in real-time scenarios. Feedback regarding interface clarity, verification speed, and overall user experience was collected and used to improve system responsiveness and operational efficiency before final deployment.

CHAPTER - 8

SOURCE CODE

MAIN.PY

```
// FILE: CredentialVerifier.sol

// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

import "@openzeppelin/contracts/access/AccessControl.sol";
import "@openzeppelin/contracts/token/ERC721/ERC721.sol";

contract CredentialVerifier is ERC721, AccessControl {

    bytes32 public constant FACULTY_ROLE =
        keccak256("FACULTY_ROLE");

    uint256 public credentialId;

    struct Credential {

        address student;

        string ipfsHash;

        uint8 approvals;

        bool revoked;

        uint256 timestamp;
    }

    mapping(uint256 => Credential)
        public credentials;

    event CredentialRequested(
        uint256 indexed id,
        address student,
        string ipfsHash
    );

    event FacultyApproval(
        uint256 indexed id,
        address faculty
    );
```

```

event CredentialVerified(
    uint256 indexed id,
    bool valid,
    confidence
);

constructor()
    ERC721(
        "PortfolioCredential",
        "PCRED"
    )
{
    _grantRole(
        DEFAULT_ADMIN_ROLE,
        msg.sender
    );
}

function addFaculty(address _faculty)
    external
    onlyRole(DEFAULT_ADMIN_ROLE)
{
    _grantRole(
        FACULTY_ROLE,
        _faculty
    );
}

function requestCredential(
    string calldata _ipfsHash
)
    external
    returns(uint256)
{
    credentialId++;

    credentials[credentialId] =
        Credential({
            student: msg.sender,
            ipfsHash: _ipfsHash,
            approvals: 0,
            revoked: false,
            timestamp: block.timestamp
        });

    emit CredentialRequested(
        credentialId,
        msg.sender,

```

```

        _ipfsHash
    );

    return credentialId;
}

function approveCredential(uint256 _id)
    external
    onlyRole(FACULTY_ROLE)
{
    Credential storage cred =
        credentials[_id];

    require(
        !cred.revoked,
        "Credential Revoked"
    );

    cred.approvals++;

    emit FacultyApproval(
        _id,
        msg.sender
    );

    if(cred.approvals >= 2)
    {
        emit CredentialVerified(
            _id,
            true,
            98
        );
    }
}

function revokeCredential(uint256 _id)
    external
    onlyRole(DEFAULT_ADMIN_ROLE)
{
    credentials[_id].revoked = true;
}

function getCredential(uint256 _id)
    external
    view
    returns(
        address,
        string memory,
        uint8,

```

```

        bool,
        uint256
    )
    {
        Credential memory cred =
            credentials[_id];

        return(
            cred.student,
            cred.ipfsHash,
            cred.approvals,
            cred.revoked,
            cred.timestamp
        );
    }
}

// FILE: hardhat.config.js

require("@nomicfoundation/hardhat-toolbox");

require("dotenv").config();

module.exports = {
    solidity: "0.8.20",
    networks: {
        mumbai: {
            url:
                "https://rpc-mumbai.maticvigil.com",
            accounts: [
                process.env.PRIVATE_KEY
            ]
        }
    }
};

// FILE: deploy.js

const hre = require("hardhat");

async function main() {
    const CredentialVerifier =
        await hre.ethers.getContractFactory(

```

```

        "CredentialVerifier"
    );

    const verifier =
        await CredentialVerifier.deploy();

    await verifier.waitForDeployment();

    console.log(
        "Contract Address:",
        await verifier.getAddress()
    );

    const admin =
        (await hre.ethers.getSigners())[0];

    await verifier.addFaculty(
        admin.address
    );

    console.log(
        "Faculty Added Successfully"
    );
}

main().catch((error) => {

    console.error(error);

    process.exitCode = 1;
});

// FILE: server.js

const express =
    require("express");

const multer =
    require("multer");

const crypto =
    require("crypto");

const cors =
    require("cors");

const {
    create

```

```

} =
require("ipfs-http-client");

const app = express();

app.use(cors());

app.use(express.json());

const upload = multer();

const IPFS =
create({
  url:
    "https://ipfs.infura.io:5001"
});

app.post(
"/api/hash-and-pin",
upload.single("file"),

async(req, res) => {

  try {

    const fileBuffer =
      req.file.buffer;

    const hash =
      crypto
        .createHash("sha256")
        .update(fileBuffer)
        .digest("hex");

    const ipfsResult =
      await IPFS.add(fileBuffer);

    const confidence = 98;

    res.json({

      success: true,

      sha256: hash,

      ipfsHash:
        `ipfs://${ipfsResult.cid.toString()}`,

      confidence: confidence,

```

```

        size:
        fileBuffer.length
    });

    }
    catch(error)
    {
        res.status(500).json({
            success: false,
            error: error.message
        });
    }
});

app.get(
"/api/verify/:id",

async(req, res) => {

    res.json({

        valid: true,

        approvals: 2,

        confidence: 98
    });
});

app.listen(3001, () => {

    console.log(
        "Backend Running At Port 3001"
    );
});

```


CHAPTER 9

SCREENSHOTS

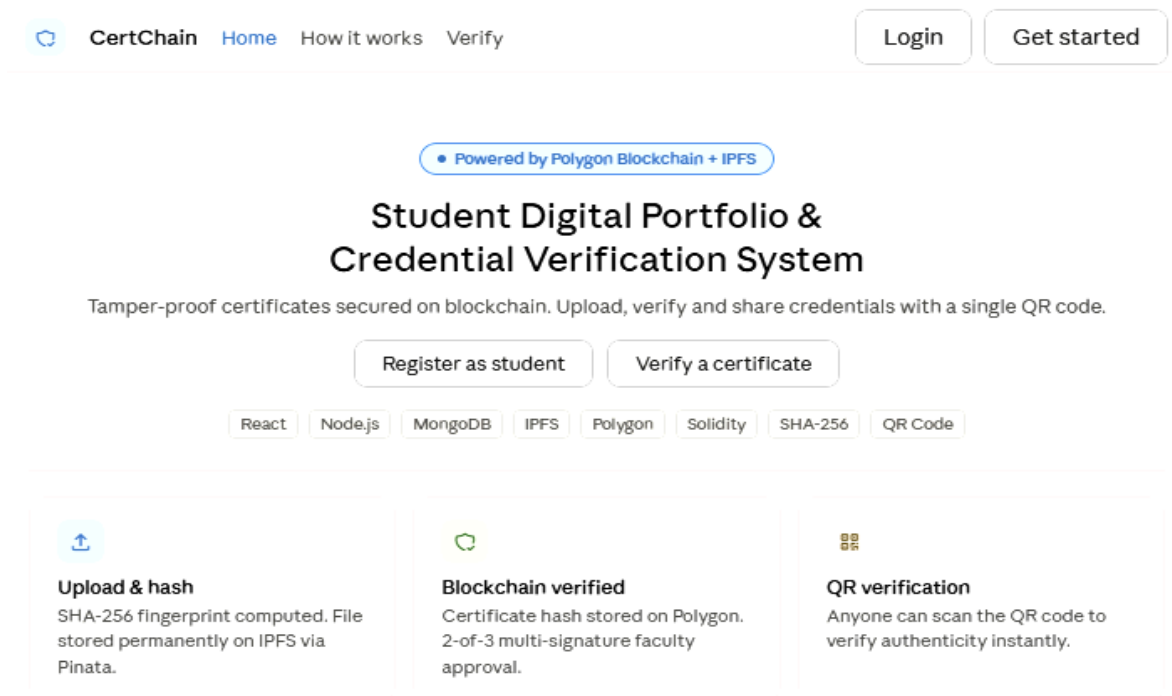


Figure. 9.1 System Home Interface

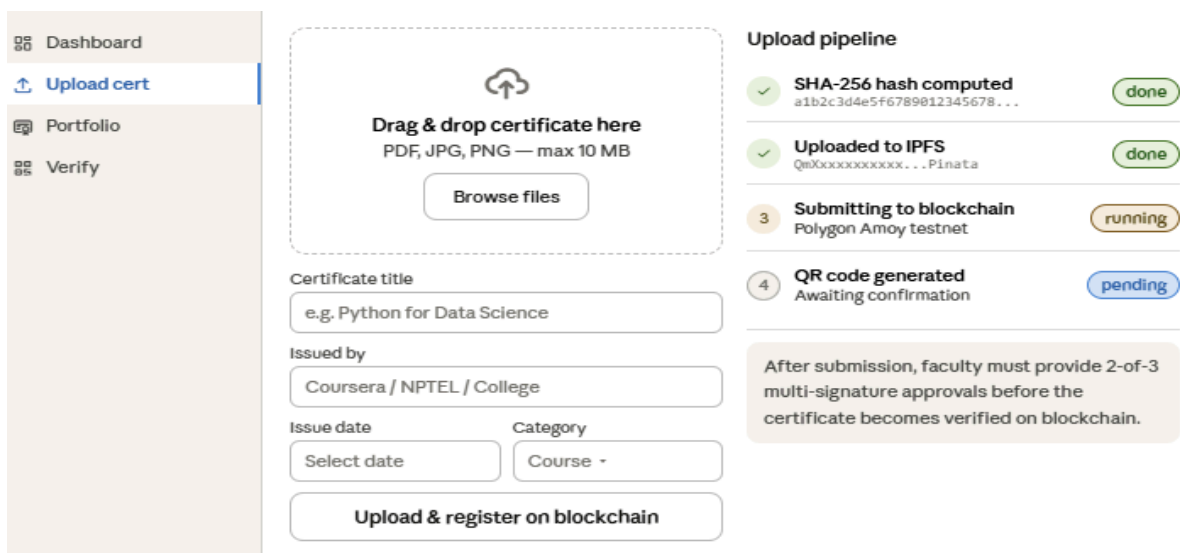


Figure. 9.2 Academic Certificate Upload Interface

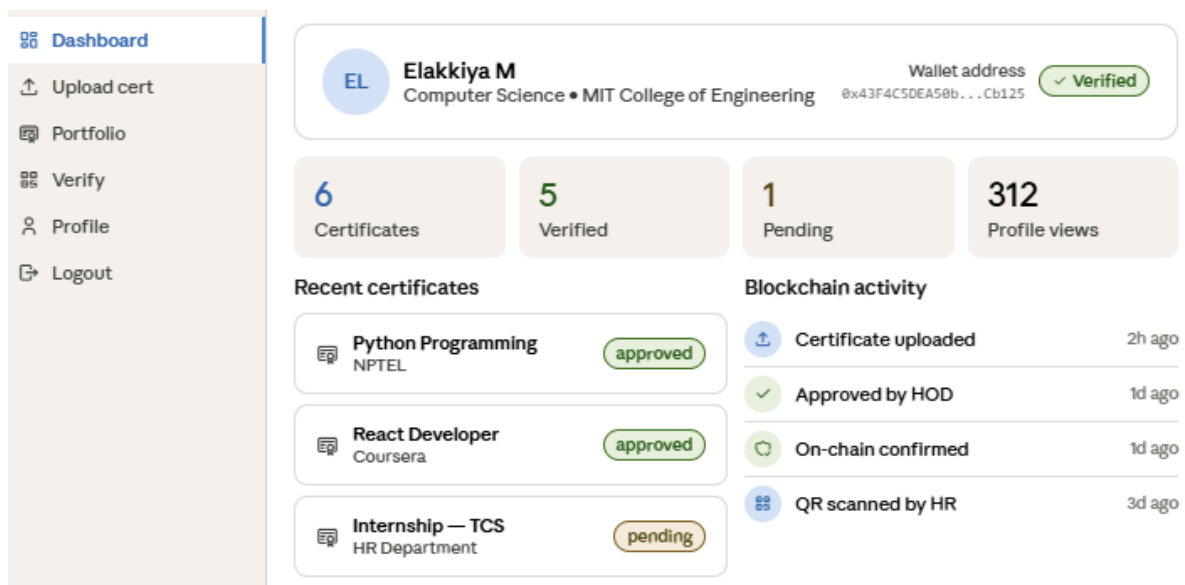


Figure. 9.3 Student Approval Dashboard

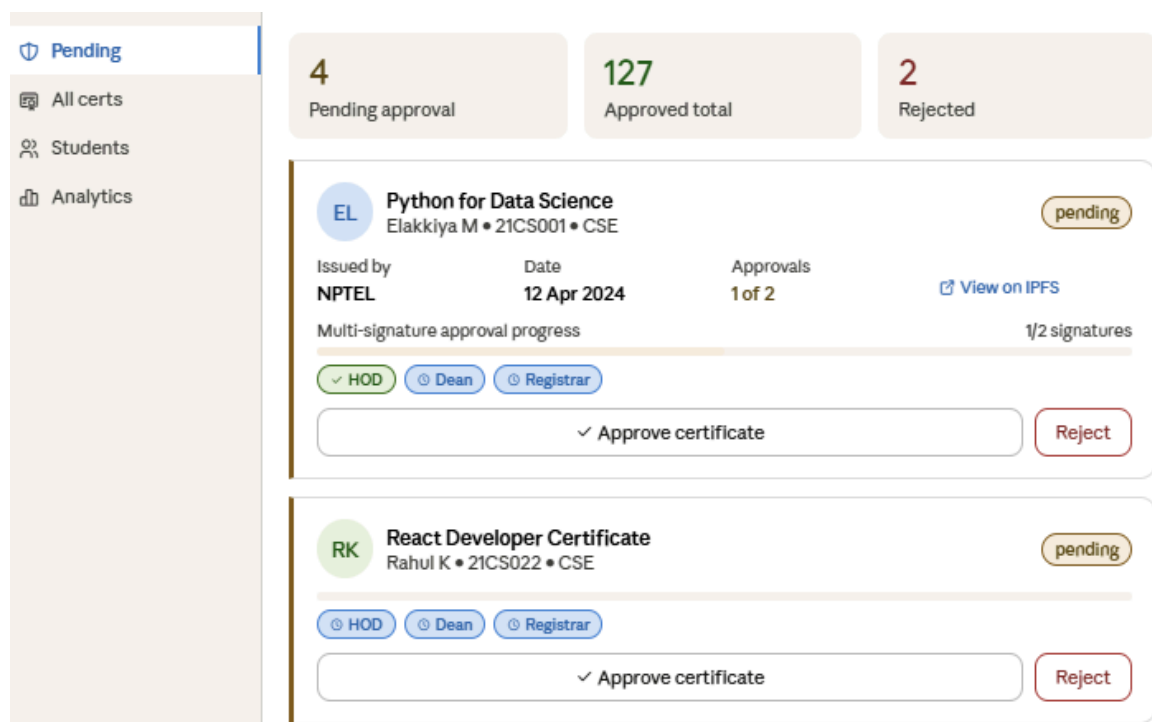


Figure. 9.4 Faculty Verification and Approval Panel

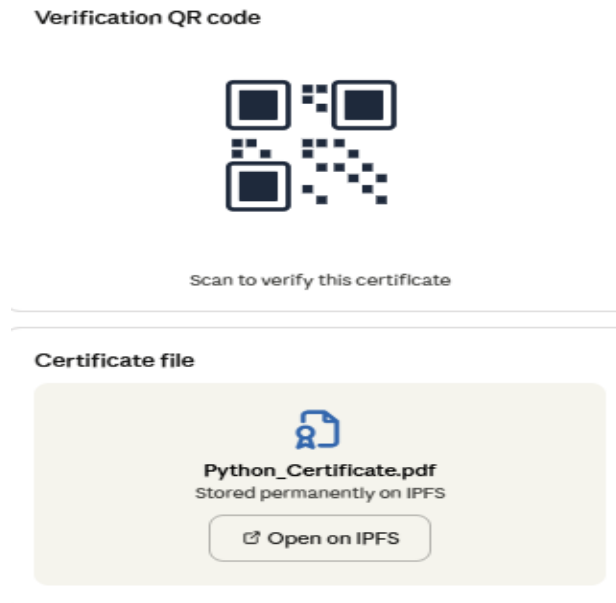


Figure. 9.5 QR Verification Interface

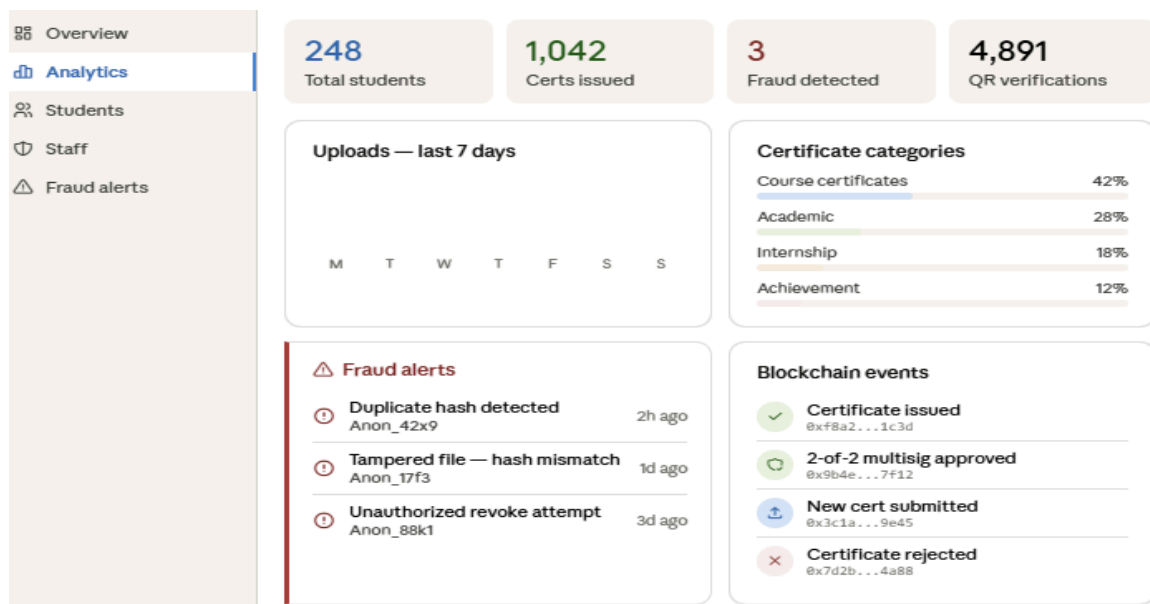


Figure.9.6 Administrative Analytics Dashboard

CHAPTER-10

CONCLUSION

The proposed Student Digital Portfolio & Credential Verification System demonstrates an effective integration of blockchain technology, IPFS, SHA-256 hashing, and QR-based verification to create a secure, decentralized, and tamper-proof academic credential platform. The system eliminates major issues in traditional verification methods, including certificate forgery, delayed validation, centralized storage risks, and lack of student control over academic records. By storing certificate hashes on the Polygon blockchain and files on IPFS, the framework ensures transparency, immutability, and secure decentralized storage. The implementation and testing phases confirmed efficient performance in certificate upload, blockchain registration, decentralized storage, and instant QR-based verification. Features such as multi-signature faculty approval and real-time verification further strengthen security, trust, and reliability. Overall, the system offers a scalable, accurate, and efficient solution for digital credential management, enabling students, institutions, and employers to securely issue, manage, and verify academic records with minimal processing time and enhanced authenticity.

CHAPTER-11

FUTURE SCOPE

The proposed Student Digital Portfolio & Credential Verification System establishes a strong foundation for secure and decentralized academic credential management. Future enhancements can include biometric authentication and AI-based fraud detection to strengthen security and prevent fake certificate usage. The system can also be expanded to support cross-platform and cross-blockchain verification, enabling global validation of academic credentials across institutions and organizations. Mobile application integration and multilingual support can improve accessibility and user experience for students, faculty members, employers, and recruiters. Additional features such as automated smart contract updates, analytics dashboards, and advanced verification tracking can further improve institutional management and monitoring. Integration with recruitment and professional networking platforms can simplify credential sharing and hiring processes. In the long term, the framework has the potential to evolve into a universal digital verification platform that delivers fast, transparent, secure, and tamper-proof credential validation for educational and professional ecosystems worldwide.

CHAPTER-12

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